

Lab-8 Ex1 - Functions Warmup (Prelab*)

This is a PreLab exercise. Please do the PreLabs after each lecture and submit before you come to lab!

As we have discussed in lecture, functions in C are quite similar to functions in maths. Let's first warm up by creating some maths function in C!

In `main.c` you see a template with two empty functions `CircleArea()` and `Circumference()`:

```
double CircleArea(double radius) {  
    // return the circle's area  
}  
  
double Circumference(double diameter) {  
    // return the circle's circumference  
}
```

The function names are clearly self-explanatory – the functions should calculate the area and circumference of a circle respectively. The first function, `CircleArea()`, will take in a parameter of `radius`, while the other, `Circumference()`, will take in parameter of `diameter`.

Assuming $\pi = 3.1416$, can you help finish the two empty functions so that the C program can have the proper behavior, as shown below? Please do NOT touch the main function!

Assume all input are valid ($\text{radius} > 0$). User-input numbers are ***bold and italic*** in the following sample runs.

Sample Run #1

Radius of circle? ***1.51***

Area: 7.1632

Circumference: 9.4876

Sample Run #2

Radius of circle? ***3***

Area: 28.2744

Circumference: 18.8496